

A Retrieval-Augmented Generation System for Automated Mind Map Creation and Enhanced Learning in Higher Education

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Abstract—The increasing reliance on AI for instant answers in higher education often leads to superficial understanding, creating a need for tools that foster deeper conceptual engagement. This paper details the analysis and design of an information system integrating Retrieval-Augmented Generation (RAG) with automated mind map visualization to enhance self-directed learning for university students. The system was developed using the Prototype model of the Software Development Life Cycle (SDLC). A quantitative approach with simple linear regression was employed to analyze data collected from a questionnaire distributed to 35 students, measuring the system's impact on learning quality. The designed system successfully transforms PDF documents into interactive, queryable mind maps. Regression analysis showed that the RAG-based system significantly impacts the quality of learning, explaining 74.6% of the variance ($R^2 = 0.746$). User evaluations and black-box testing confirmed the system's functionality and effectiveness in enhancing conceptual understanding and providing a valuable, interactive educational tool. In conclusion, the proposed system demonstrates strong potential to support deeper learning and can serve as a foundation for future educational technology innovations.

Keywords—quality education, retrieval-augmented generation, mind map, information systems, higher education

I. INTRODUCTION

Higher education institutions face the ongoing challenge of adapting to new ways students access and comprehend digital learning materials. The shift from conventional to online learning has driven universities to optimize their use of information technology, aiming for a more flexible, efficient, and student-centered educational experience. This trend requires information systems that move beyond presenting static information to facilitating a deep, interactive understanding of complex subjects tailored to individual learners [1].

To meet this demand, recent technological advancements offer powerful new tools. One of the most prominent is Retrieval-Augmented Generation (RAG), an AI architecture that grounds Large Language Models (LLMs) in external

knowledge sources to provide factual, context-aware responses. The state-of-the-art in RAG has moved beyond simple text retrieval, now incorporating multimodal data and agentic workflows that dramatically enhance accuracy and relevance in educational settings [2]. In parallel, AI-powered visualization tools have transformed how information is structured and absorbed. Modern mind mapping platforms, for instance, leverage deep learning to automatically extract and hierarchically organize key concepts from complex documents [3]. Meta-analyses confirm that such visual learning tools significantly improve comprehension, retention, and student motivation across diverse disciplines [4]. While the impact of LLMs on education is profound, enabling everything from personalized tutors to automated curriculum design [5]. A major research gap remains in integrating these advanced technologies. Most current RAG applications focus on text-based interaction, failing to incorporate innovative learning visualizations that cater to different cognitive styles.

One of the most promising technological approaches in this context is Retrieval-Augmented Generation (RAG), an AI architecture that combines information retrieval from external sources with the contextual answer generation capabilities of Large Language Models (LLMs) [6]. RAG enhances the learning experience by providing relevant, source-based responses, making the process more efficient and adaptive [7]. However, the effectiveness of a learning system is determined not just by the quality of its information but also by its presentation [8]. Mind maps, a visual tool for structuring information hierarchically, have been proven to improve student comprehension and memory by illustrating the connections between concepts [3]. This visual approach helps in reducing the cognitive load and making complex information more digestible [9]. The automated generation of these mind maps from text can further enhance learning by saving time and providing a consistent structure for study materials [10]. While RAG offers significant improvements in accuracy and context, a major research gap remains in integrating RAG with innovative learning visualizations. Most existing RAG applications in education focus on text-

based interactions, which may not be optimal for all learning styles.

This research presents the analysis and design of VizMind AI, an information system that integrates a RAG architecture with the automated generation of mind maps from academic documents. The system was developed using a Prototype model to ensure user-centric design through iterative feedback. It is designed to address a critical gap in modern digital learning: the tendency for students to rely on instant, text-based answers from AI without developing a deeper, conceptual understanding. VizMind AI allows students to upload learning materials in PDF format, which the system then processes to extract key information, perform contextual searches, and generate both a textual answer and a visual mind map representing the document's structure. This dual approach supports a more profound, self-directed learning process, particularly for visual learners.

The primary objective of this research is to analyze and design a system that leverages RAG and mind map visualization to improve the quality of learning in higher education, focusing on key indicators such as the availability of structured learning resources, the ability to understand material independently, and the efficiency of academic inquiry. From a technological perspective, the scope of this paper is focused on the analysis, design, and implementation of an information system architecture. This research contributes to the fields of software engineering and educational technology by presenting a full-stack application that synergistically combines a RAG-based AI workflow with an interactive visualization front-end. We detail the system architecture, the AI processing pipelines built with LangGraph, and the integration of external services, providing a practical blueprint for developing similar intelligent learning systems.

II. LITERATURE REVIEW

A. Retrieval-Augmented Generation (RAG)

Retrieval-Augmented Generation (RAG) is an advanced technique for Large Language Models (LLMs) that combines document retrieval from an external knowledge base with a model's generative capabilities. It uses a retriever model to find relevant external information, which an LLM like BERT or GPT then uses to construct a response [11]. By grounding the model in external documents, this approach crucially reduces errors and hallucinations. Unlike standard models that rely only on their training data, RAG systems retrieve relevant documents to use as context, producing more accurate and traceable answers ideal for educational settings [12].

For instance, a study on a RAG system called OwlMentor found that while a direct correlation to learning gains was not established, users showed higher gains than non-users [13]. Similarly, a RAG chatbot in an R programming course saw higher use among students with prior knowledge and received high overall satisfaction [14]. Furthermore, a meta-analysis confirmed the strong positive impact of concept maps on academic achievement [15]. These findings collectively support integrating RAG and concept mapping to improve learning outcomes in higher education.

B. Automated Mind Map Visualization

Visual representations like mind maps are critical for structuring information to enhance comprehension and

memory [16]. By presenting concepts graphically, they help users map relationships between ideas, simplifying complex documents into easier-to-understand elements [17]. This clear structure supports knowledge retention, and studies show that the systematic use of mind maps enhances learning outcomes, particularly students' ability to form connections between concepts [18]. Modern applications include using knowledge graphs to prompt a "graph of thoughts" in LLMs, improving their reasoning and transparency [19].

C. Enhanced Learning in Higher Education

In modern education, the quality of learning is not only determined by the availability of material but also by how technology can be used to improve the effectiveness of information delivery [20]. The application of RAG in intelligent education systems allows for an enhanced learning experience by providing answers that are more contextual, accurate, and relevant to students' needs. In addition to providing relevant information, RAG-based systems also enhance students' ability to understand material independently [21]. Such systems can help students better understand concepts without direct guidance from lecturers because the information provided is concise, contextual, and easy to comprehend, which is crucial for self-directed learning environments [22]. Furthermore, system efficiency, measured through low latency and the speed of generating relevant answers, supports a quick response to questions or assignments, making the learning process more active and unhindered by delays in obtaining information [15].

D. Research Gap

While Retrieval-Augmented Generation (RAG) improves AI answer accuracy [21] and mind maps are effective visualization tools [15], there is a research gap in integrating both into a unified, interactive learning system. Current applications are typically limited to text-based RAG interactions [14] or static mind map generation [3].

This study introduces VizMind AI, a system that fills this gap by transforming the mind map into an interactive interface for knowledge exploration. It allows students to engage in a context-aware, RAG-powered Q&A session directly on mind map nodes, enabling deep inquiry without losing conceptual context.

Our approach addresses passive learning by allowing students to actively interrogate visual knowledge structures. This is supported by findings that integrating mind mapping with a generative AI chatbot significantly improves learning performance compared to using a chatbot alone [10], meeting a critical need in modern higher education.

III. RESEARCH METHODOLOGY

A. Research Approach

This study uses a quantitative approach with simple linear regression analysis to measure the impact of the RAG-based system on student learning quality. This method was chosen to empirically test the hypotheses regarding the influence of system features, such as the quality of information retrieval and answer generation, on specific learning outcomes.

B. Development Methodology

The system was developed using the Prototype model of the Software Development Life Cycle (SDLC). This iterative approach was selected because it allows for early user feedback and refinement, which is critical for developing a

user-centric educational tool that integrates advanced technologies like Retrieval-Augmented Generation (RAG) and Large Language Models (LLMs) [23]. The process involved identifying initial requirements, creating a functional prototype, gathering user feedback through testing sessions, and iteratively refining the system based on that feedback.

C. Data Collection and Analysis

Data was collected through a questionnaire distributed to 35 university students from various disciplines and semester levels. Although the sample size may appear limited, it is considered sufficient for an initial prototype-based investigation. At this stage, the primary objective is to obtain focused feedback to inform system refinement and to evaluate a core hypothesis, rather than to produce findings that can be generalized to a wider population. In this context, a smaller, focused group is highly effective for the iterative feedback loop required by the Prototype model, ensuring the system's features are robust and user-centric before a wider deployment. The demographic diversity across disciplines and semester levels provides a representative snapshot of the target user group [24].

The validity test was conducted using the Pearson Product-Moment correlation coefficient, with an R-value greater than the R-table and a significance value of less than 0.05, confirming that the questions accurately measured the intended variables.

The decision to use a RAG architecture is validated by recent benchmarks showing that advanced RAG implementations significantly outperform baseline LLMs, with accuracies reaching up to 76.74% on complex reasoning tasks [25]. We adopted a similar retrieval-augmented approach enhanced with mind map generation and, through regression analysis, found a significant positive relationship between system accuracy and student learning outcomes. This combination of high benchmark performance and our own findings confirms that the system's technical foundation is robust and capable of improving learning in higher education.

TABLE I. ACCURACY (%) OF LLAMA3.3-70B MODELS ON THE MATHQA DATASET

Methods	LLaMA3.3-70B
Baseline	69.07
RAG	71.16
RAG+	71.86
AFRAG	70.23
AFRAG+	71.86
GraphRAG	68.37
GraphRAG+	69.00
Rerank RAG	74.65
Rerank RAG+	76.74

IV. RESULT AND DISCUSSION

A. Initial Gathering

The initial user evaluation of the prototype yielded critical insights for system improvement. Feedback from user personas highlighted a need for more context and

interactivity. Key suggestions included adding descriptive summaries to each sub-topic in the mind map and enabling an interactive Q&A feature directly on the nodes, allowing users to query specific concepts without leaving the visual interface. These recommendations were incorporated into the final system, which now features a fully interactive mind map, a contextual Q&A panel for each node, and a more intuitive interface for document management. Subsequent black-box testing confirmed that all features of the final system, including user login, file upload, mind map generation, and the interactive Q&A functionality, performed as expected and met the specified functional requirements.

B. Regression Analysis Findings

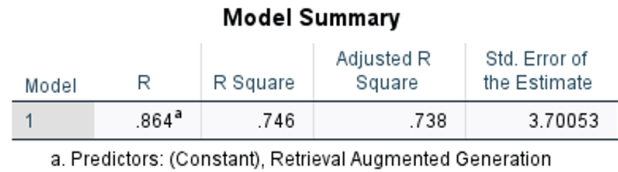


Fig. 1. Regression model summary.

The core of our quantitative findings stems from the regression analysis, which demonstrated that Retrieval-Augmented Generation has a significant positive effect on the quality of learning, explaining 74.6% of the variance (R Square = 0.746). This strong correlation underscores the system's effectiveness.

These findings confirm the central hypothesis that a RAG-based system enhanced with mind map visualization is an effective tool for modern higher education. The system successfully addresses the initial problems of passive learning and lack of contextual understanding by providing a more interactive, visual, and in-depth learning experience. The strong statistical correlation found in our results highlights the value of integrating advanced AI techniques with user-centric design in educational technology.

C. Evaluation Metric

We evaluated our system's effectiveness using two key measures: its technical accuracy and its impact on student learning.

First, the choice of a Retrieval-Augmented Generation (RAG) architecture is supported by external benchmarks. As shown in Table I, advanced RAG models achieve high accuracy on complex datasets, with top models reaching 76.74% [25]. This confirms that the underlying technology is technically sound and highly capable.

Second, we measured the system's educational impact through our user study. The regression analysis revealed that our system accounts for 74.6% of the variance (R² = 0.746) in the quality of student learning. This strong statistical result demonstrates that the system's features—such as the quality of its information retrieval and the clarity of its answers—are directly linked to improved learning outcomes.

In conclusion, these findings show that the system's strong technical performance translates into a significant and measurable positive impact on the student learning experience, validating our research approach.

D. System Architecture

The architecture of VizMind AI is designed as a full-stack application with a clear separation of concerns between the frontend, backend, AI processing, and data layers. This modular design ensures scalability, maintainability, and security [26].

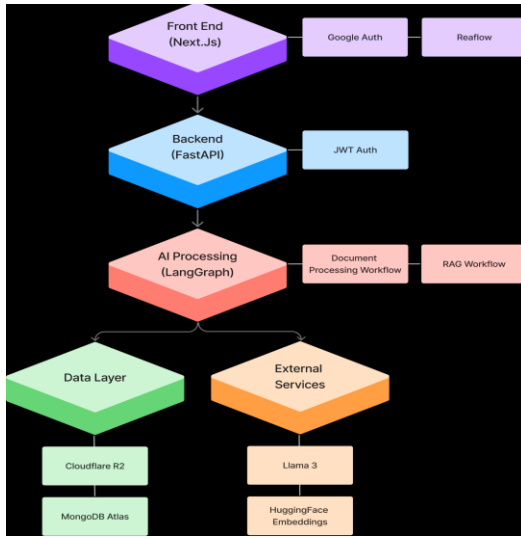


Fig. 2. System architecture of VizMind AI.

a) *Frontend*: The frontend is responsible for the user interface and experience. It includes components for user authentication, a drag-and-drop interface for file uploads, and an interactive canvas for rendering the mind map.

b) *Backend*: The backend serves as the central hub for the application, handling API requests, user authentication with JWT tokens, and business logic. It leverages FastAPI's high performance and asynchronous capabilities.

c) *AI Processing*: The core of the system's intelligence lies in the AI processing layer, built using LangGraph. This layer contains two main workflows: a document processing pipeline that transforms PDFs into structured data and a RAG pipeline for answering user queries.

d) *Data Layer*: The data layer consists of a MongoDB Atlas database for storing user data, mind map structures, and document chunks with their vector embeddings. Cloudflare R2, an S3-compatible object storage, is used for storing the uploaded PDF documents.

e) *External Services*: The system integrates with several external services to provide its core functionalities. These include Google OAuth for secure user authentication, the Groq API for high-speed access to the Llama 3 Large Language Model for generative tasks, and HuggingFace for providing the sentence-transformer models used to create vector embeddings.

E. AI Workflow and Implementation

The VizMind AI system is powered by two primary LangGraph workflows: the Document Processing Pipeline and the Retrieval-Augmented Generation (RAG) Pipeline. These workflows are designed to be robust, with built-in error handling and retry mechanisms [27].

a) *Document Processing Pipeline*: The document processing pipeline is a multi-stage workflow that transforms an unstructured PDF document into a structured, queryable format.

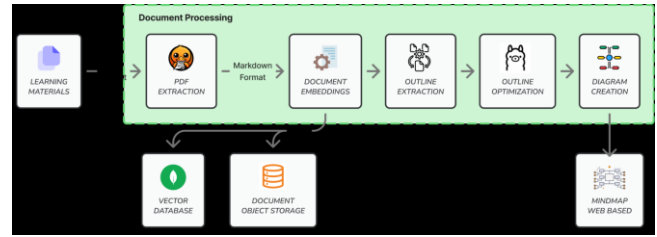


Fig. 3. Document processing pipeline.

The process begins with Content Extraction, for which we selected Docling, an open-source package powered by specialized AI models for layout analysis and table structure recognition [28]. When a user uploads a PDF, the system ingests the file and converts the document into a clean Markdown format that retains structural elements like headings and tables.

Following extraction, the system employs a sophisticated two-stage process to generate the mind map: Outline Extraction and Mind Map Optimization. First, the raw Markdown is passed to an Extract Outline node, which divides the content into smaller sections and processes them in parallel to generate a foundational outline of the document's key topics. This parallel processing architecture significantly enhances performance and speed.

Next, this raw outline is sent to a dedicated Optimize Outline node. This crucial step refines the structure by applying several mind mapping best practices, including removing duplicate concepts, ensuring terminological consistency, balancing the hierarchy for visual clarity, and creating concise, keyword-focused labels. This optimization transforms the initial outline into a logically grouped and structurally sound framework, which is essential for creating an effective visual learning tool. Only after this refinement is the final hierarchical JSON structure generated, which is required by the frontend's Reaflow library for rendering.

The final stage of document processing is Semantic Chunking and Embedding. The cleaned Markdown is split into smaller, semantically meaningful chunks, which are then converted into vector embeddings for the RAG pipeline. The system splits the document based on its headings to ensure that the chunks are contextually coherent. Each chunk is then converted into a 768-dimensional vector embedding using the *sentence-transformers/paraphrase-multilingual-mpnet-base-v2* model and stored in MongoDB Atlas with a vector search index.

b) *Retrieval-Augmented Generation (RAG) Workflow*: The RAG workflow is triggered when a user asks a question about a concept in the mind map. The first step is Document Retrieval, where the system retrieves the most relevant document chunks from MongoDB Atlas based on the user's query. This provides the LLM with the necessary context from the user's document to generate an accurate and grounded answer. The user's query is converted into a vector embedding, and a cosine similarity search is performed on the MongoDB collection.

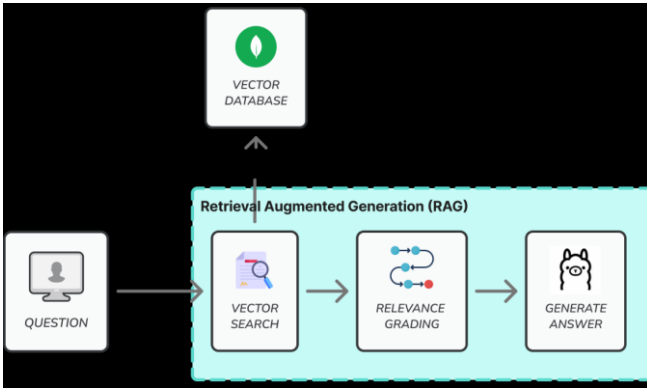


Fig. 4. Retrieval-augmented generation pipeline.

The retrieved chunks then undergo Relevance Grading. This step acts as a filter to remove any irrelevant chunks, improving the quality of the context provided to the LLM. Each retrieved document is passed to the LLM with the following prompt to get a binary "yes" or "no" score on its relevance to the user's question.

Finally, Answer Generation occurs. The LLM synthesizes information from the filtered, relevant document chunks to provide a comprehensive answer. The relevant chunks are combined into a single context string and passed to the LLM along with the user's original question and a detailed prompt that instructs the model to answer based only on the provided context, be comprehensive yet concise, and structure its response in Markdown. The generated answer is then returned to the user with citations for the source documents.

F. User Interface and Experience

The frontend of VizMind AI is designed to be intuitive, responsive, and engaging, providing a seamless experience for users.

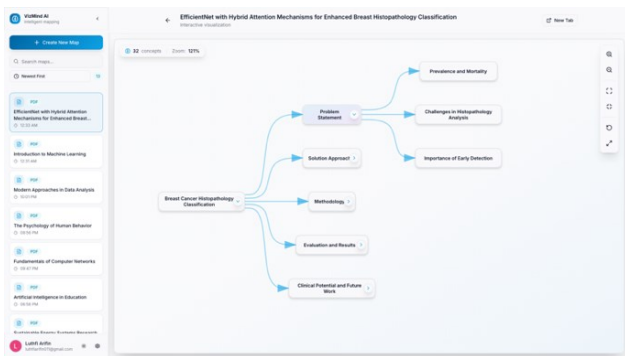


Fig. 5. Mind map user interface.

The Mind Map User Interface is rendered using the Reaflow library, which provides a highly interactive and customizable canvas. Users can easily pan, zoom, and explore the visually organized concepts derived from their uploaded documents. The hierarchical structure and clear labeling of nodes facilitate a quick understanding of the document's main topics and their interconnections.

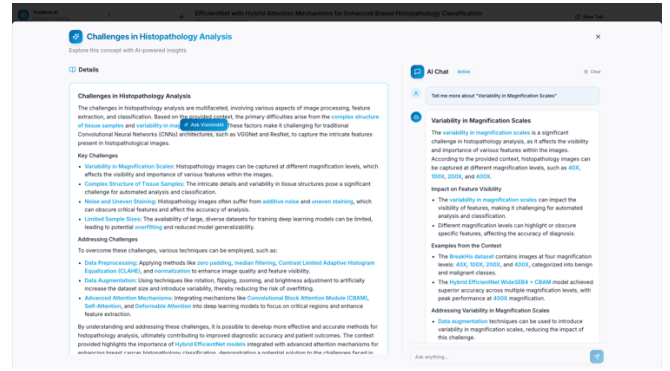


Fig. 6. Node detail user interface.

When a user interacts with a specific node on the mind map, the Node Details UI is displayed. This panel opens to provide more detailed information about the selected concept, including an initial overview generated by the RAG pipeline. Within this panel, users can engage in a chat interface to ask follow-up questions related specifically to that node. The conversation history is maintained for each node, allowing for a continuous and context-aware dialogue and deeper exploration of individual topics without losing the overall visual context of the mind map.

G. Discussion

The strong statistical correlation between the system's features and improved learning outcomes demonstrates the value of integrating advanced AI techniques with user-centric design in educational technology. Our primary finding, that the RAG-based system explains 74.6% of the variance in learning quality empirically supports other research identifying RAG as a pivotal technology for education, largely due to its ability to ground LLM-generated content in factual sources.

The system's success in improving the availability of learning resources aligns with findings that report RAG-based chatbots enhance the learning experience by providing contextual and source-based responses [21]. Moreover, by integrating RAG with automated mind map visualization, our system addresses a key gap identified in the literature. While some studies have confirmed the value of AI-generated mind maps for knowledge retention [3], our research demonstrates that the combination of retrieval-augmented generation and visual structuring provides a synergistic learning tool. This is consistent with other research which found that integrating mind mapping with a generative AI chatbot led to significantly better learning performance compared to using a chatbot alone [10]. Our system provides a practical implementation of this synergy, transforming passive text into searchable interactive knowledge maps.

Despite these positive results, it is important to acknowledge the limitations of this study. The evaluation was conducted with a sample of 35 students, which, while sufficient for initial validation within the Prototype model, may not be large enough to generalize the findings to the broader student population [24]. The study also relied on a simulation of a learning environment rather than a long-term, in-classroom deployment, which would be necessary to measure the system's impact on actual academic outcomes.

The effectiveness of the system is also inherently tied to the capabilities of the underlying Large Language Model (Llama 3) and the Docling extraction tool. Although RAG is

designed to mitigate the risk of "hallucinations" by grounding the model in provided documents, it does not eliminate it entirely. The quality of the output is highly dependent on the clarity and structure of the source PDF, and the system may be less effective with poorly formatted or scanned documents.

The design of the proposed system provides a concrete solution to the challenges of passive learning and lack of conceptual depth in higher education. By combining Retrieval-Augmented Generation with automated mind map visualization, the system not only delivers accurate and contextual responses but also transforms them into structured visual knowledge representations. This dual approach allows students to engage with materials more interactively, improving both comprehension and retention while reducing cognitive overload from unstructured text. The regression analysis results, showing that the system explains 74.6% of the variance in learning quality, further validate its effectiveness as a tool that fosters independent, deeper, and more efficient learning.

V. CONCLUSION AND FUTURE WORK

This paper presented the analysis and design of an information system that integrates Retrieval-Augmented Generation (RAG) with automated mind map visualization to improve learning quality in higher education. The developed system, VizMind AI, demonstrates a significant positive impact on learning by addressing the problems of passive learning and lack of deep conceptual understanding. Regression analysis confirmed that the system is highly effective, with RAG explaining 74.6% of the variance in learning quality. This finding is further supported by external benchmarks demonstrating the high technical proficiency of the underlying RAG architecture, which has shown accuracy rates up to 76.74% on complex knowledge-intensive tasks [25]. Furthermore, specific attributes of the system—such as the quality of information retrieval, the contextual accuracy of generated answers, and overall system efficiency—were shown to directly enhance the availability of structured resources, students' ability to understand material independently, and the speed of academic inquiry. Through iterative prototype development and user-centered design, VizMind AI has proven to be a robust and practical solution that supports adaptive, interactive, and student-centered learning in modern higher education. This work contributes a practical tool for students, who are the primary beneficiaries of this technology. VizMind AI empowers learners to independently process and understand academic materials through structured visualizations and contextual responses, reducing their reliance on lecturers or external tutoring. By transforming complex documents into interactive knowledge maps, the system supports deeper engagement, improves retention, and enhances critical thinking. This contribution is particularly valuable in higher education, where self-directed learning is essential for academic success and lifelong learning skills.

Future work will focus on extending the system's capabilities. We plan to incorporate additional models for classifying figures, recognizing equations, and identifying code blocks to further enrich the extracted document metadata. We also aim to improve GPU acceleration support for the AI models to enhance performance. Further user studies will be conducted to evaluate the long-term impact of the system on student learning outcomes and to gather

feedback for continuous improvement. We also plan to expand the system to support a wider range of document formats beyond PDF and to explore collaborative features that would allow multiple users to interact with the same mind map.

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Dataset Availability Statement: The dataset used in this study is publicly available on Github and can be accessed at: <https://github.com/edu-rag/vizmind-ai>

REFERENCES

- [1] V. Mirata, F. Hirt, P. Bergamin, and C. van der Westhuizen, "Challenges and contexts in establishing adaptive learning in higher education: findings from a Delphi study," *International Journal of Educational Technology in Higher Education*, vol. 17, no. 1, Dec. 2020, doi: 10.1186/s41239-020-00209-y.
- [2] A. J. Oche, A. G. Folashade, T. Ghosal, and A. Biswas, "A systematic review of key retrieval-augmented generation (RAG) systems: progress, gaps, and future directions," Jul. 2025, [Online]. Available: <http://arxiv.org/abs/2507.18910>
- [3] K. Joseph, A. Gaiwal, P. Raikar, G. Raorane, and T. Rokade, "Article information Article title AI-augmented mind maps: leveraging large language models and visualization tools for enhanced knowledge representation uses gemini API for initial topic mapping and LLaMA-3 for deeper expansion automatically generates editable mind maps from academic pdfs leverages react flow, flask, and mongodb for scalable and dynamic visualization," May 2025, [Online]. Available: <https://ssrn.com/abstract=5271381>
- [4] J. Schoenherr, A. R. Strohmaier, and S. Schukajlow, "Learning with visualizations helps: A meta-analysis of visualization interventions in mathematics education," Nov. 01, 2024, Elsevier Ltd. doi: 10.1016/j.edurev.2024.100639.
- [5] R. Beale, "The revolution has arrived: what the current state of large language models in education implies for the future," Jul. 2025, [Online]. Available: <http://arxiv.org/abs/2507.02180>
- [6] J. Martinez-Romo, L. Araujo, L. Plaza, and F. López-Ostenero, "Generative AI for education: a retrieval-augmented system for effective feedback in self-assessment," in 2025 IEEE Global Engineering Education Conference (EDUCON), 2025, pp. 1–9. doi: 10.1109/EDUCON62633.2025.11016446.
- [7] Z. Li, Z. Wang, W. Wang, K. Hung, H. Xie, and F. L. Wang, "Retrieval-augmented generation for educational application: A systematic survey," Jun. 01, 2025, Elsevier B.V. doi: 10.1016/j.caeai.2025.100417.
- [8] Y. Gao *et al.*, "Retrieval-augmented generation for large language models: a survey," Mar. 2024, [Online]. Available: <http://arxiv.org/abs/2312.10997>
- [9] Lazaro. Iliadis, Serafin. Alonso, and L. Iliadis, *Engineering Applications of Neural Networks*. 2023.
- [10] X. Ye, W. Zhang, Y. Zhou, X. Li, and Q. Zhou, "Improving students' programming performance: an integrated mind mapping and generative AI chatbot learning approach," *Humanit Soc Sci Commun*, vol. 12, no. 1, Dec. 2025, doi: 10.1057/s41599-025-04846-4.
- [11] P. Lewis *et al.*, "Retrieval-augmented generation for knowledge-intensive nlp tasks," 2020. [Online]. Available: <https://github.com/huggingface/transformers/blob/master/>
- [12] M. Hindi, L. Mohammed, O. Maaz, and A. Alwarafy, "Enhancing the precision and interpretability of retrieval-augmented

- generation (RAG) in legal technology: a survey,” IEEE Access, vol. 13, pp. 46171–46189, 2025, doi: 10.1109/ACCESS.2025.3550145.
- [13] D. Thüs, S. Malone, and R. Brünken, “Exploring generative AI in higher education: a RAG system to enhance student engagement with scientific literature,” *Front Psychol*, vol. 15, 2024, doi: 10.3389/fpsyg.2024.1474892.
 - [14] G. Lang and T. Gurpinar, “AI-powered learning support: a study of retrieval-augmented generation (RAG) chatbot effectiveness in an online course,” *Information Systems Education Journal*, vol. 23, no. 2, pp. 4–13, 2025, doi: 10.62273/zklk5988.
 - [15] E. Izci and E. A. Akkoc, “The impact of concept maps on academic achievement: a meta-analysis,” *Heliyon*, vol. 10, no. 1, Jan. 2024, doi: 10.1016/j.heliyon.2023.e23290.
 - [16] Y. Xue, H. Mu, L. Xue, and X. Wang, “Teaching innovation and practice of mind mapping applied to engineering drawing course,” in *2023 IEEE 12th International Conference on Educational and Information Technology (ICEIT)*, 2023, pp. 156–161. doi: 10.1109/ICEIT57125.2023.10107785.
 - [17] T. J. Chen, R. R. Mohanty, and V. R. Krishnamurthy, “Queries and cues: textual stimuli for reflective thinking in digital mind-mapping,” *Journal of Mechanical Design*, vol. 144, no. 2, 2022, doi: 10.1115/1.4052297.
 - [18] H. Aljamal *et al.*, “Efficacy of mind maps and concept maps in enhancing academic performance among undergraduate medical students in the preclinical stage: a systematic review,” 2025, Springer Science and Business Media B.V. doi: 10.1007/s10459-025-10437-4.
 - [19] Y. Wen, Z. Wang, and J. Sun, “MindMap: knowledge graph prompting sparks graph of thoughts in large language models,” Mar. 2024, [Online]. Available: <http://arxiv.org/abs/2308.09729>
 - [20] A. I. Hatamleh and I. M. Hatamleh, “The importance of using ICT in improving the quality of education,” in *2024 ASU International Conference in Emerging Technologies for Sustainability and Intelligent Systems (ICETISIS)*, 2024, pp. 95–99. doi: 10.1109/ICETISIS61505.2024.10459655.
 - [21] J. Swacha and M. Gracel, “Retrieval-augmented generation (RAG) chatbots for education: a survey of applications,” Apr. 01, 2025, Multidisciplinary Digital Publishing Institute (MDPI). doi: 10.3390/app15084234.
 - [22] H. Soliman, N. Pengel, N. Duong-Trung, H. Kotte, and M. Kravčík, “Retrieval-augmented chatbots for scalable educational support in higher education milos kravcik deutsches forschungszentrum für künstliche intelligenz retrieval-augmented chatbots for scalable educational support in higher education,” 2025. [Online]. Available: <https://www.researchgate.net/publication/389582355>
 - [23] E. Bjarnason, F. Lang, and A. Mjöberg, “An empirically based model of software prototyping: a mapping study and a multi-case study,” *Empir Softw Eng*, vol. 28, no. 5, Sep. 2023, doi: 10.1007/s10664-023-10331-w.
 - [24] G. A. Johanson and G. P. Brooks, “Initial scale development: sample size for pilot studies,” *Educ Psychol Meas*, vol. 70, no. 3, pp. 394–400, 2010, doi: 10.1177/0013164409355692.
 - [25] Y. Wang *et al.*, “RAG+: enhancing retrieval-augmented generation with application-aware reasoning,” Sep. 2025, [Online]. Available: <http://arxiv.org/abs/2506.11555>
 - [26] T. Zheng and J. Fu, “Design of medical knowledge question and answer information system based on RAG,” 2025, pp. 175–180. doi: 10.2991/978-94-6463-742-7_20.
 - [27] R. Shan, “OpenRAG: open-source retrieval-augmented generation architecture for personalized learning,” in *2024 4th International Conference on Artificial Intelligence, Robotics, and Communication (ICAIRC)*, 2024, pp. 212–216. doi: 10.1109/ICAIRC64177.2024.10900069.
 - [28] C. Auer *et al.*, “Docling technical report,” Dec. 2024, [Online]. Available: <http://arxiv.org/abs/2408.09869>